- The **pupil** or black dot at the center of the eye is a hole through which light can enter the eye.
- The **iris** or coloured part, surrounding the pupil. It controls how much light enters the eye by changing the size of the pupil.
- The **cornea**, a clear window at the front of the eye covers the iris and pupil.
- A clear **lens**, located behind the pupil, acts like a camera lens by focusing light in to the retina at the back of the eye.
- The **retina** is a light sensitive inner lining at the back of the eye. Ten different layers of cells work together in the retina to detect light and turn it into electrical impulses.
- The retina has special cells called **cones** and **rods** are located in the retina. These all are known as photoreceptors and help absorb light.
- Cones cone cells help us see **colour** and detail.
- Rods rod cells allow us to see in poor lighting and give us our **night vision**.

Functions [Physiology] of the Eye

1. Light reflects of the object we are looking at
2. Light rays enter the eye through the cornea at the front of the eye.
3. The light passes through a watery fluid and enters the pupil to reach the lens.
4. The lens can change in thickness to bend the light, which will focus it on to the retina at the back of the eye.
5. On the way to the retina, light passes through a thick, clear fluid called a vitreous humour. The vitreous humor fills the eyeball and helps maintain its round shape.
6. The light then reaches the back of the eyes and hits the retina. The retina translates the light into electrical impulses which are then carried to the brain by the optic nerve.

Finally, the visual cortex of the brain interprets these impulses at what we see.



ABOUT EYE BANK

In India, we have an estimated 4.6 million people with corneal blindness that is curable through corneal transplantation made possible by eye donation. The cornea is the transparent outer portion of the eyeball that transmits light to the retina. It is a 1/2-inch wide film of tissue that forms a protective covering on the front of the eye. Corneal blindness can be cured in many cases through the transplant of a donated cornea. In 1905 Eduard Konrad Zirm, MD, performed the first successful corneal transplant. In 1944 Paton established the world's first eye bank, the Eye-Bank for Sight Restoration, in New York. Certified Eye Bank Technicians are re-certified every 3 years. **Anyone can be an eye donor!** Cornea donation is not dependent on age, eye colour, blood type or even eye sight quality. The eyes have to be removed within six hours of death. Cornea transplant is usually performed within 4 days after donation, depending upon the method of cornea preservation. Eye donation gives sight to two blind persons. One blind person is given one eye. Almost anyone between the ages of 2 and 75 can be an eye donor, regardless of poor vision, diabetes, or cancer. In fact, only under the following conditions can eye donation definitely not occur: HIV or AIDS, Active Hepatitis, Active Syphilis, Rabies, Viral Encephalitis, Leukemia, Active Lymphoma, or Active Meningitis. It is illegal to buy and sell human eyes, organs and tissues. Any cost associated with eye procurement are absorbed by the eye bank.

MAKE EYE DONATION A FAMILY TRADITION

Your eyes after death need not perish. Help blinds to see, donate eyes after death.

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